

EMABE • PGDBA • SESSION-10 PROJECT

What drives the batter's performance in a season?

A panel-data study of IPL batting (2008–2026), built from Cricsheet ball-by-ball data

Submitted by: _____

Instructor: _____

Methods: Pooled OLS • Fixed & Random Effects • Hausman • Breusch–Pagan LM • Arellano–Bond GMM

MOTIVATION

The question — and why a panel?

We watch the same batters season after season, so the right tool is panel data (Sessions 8 & 9).

A batter's output each season depends partly on **who he is** (talent) and partly on **what changed that year**. A panel lets us separate the two — an ordinary regression cannot.

The model we estimate

$$\text{Runs}_{it} = \alpha_i + \beta_1 \cdot \text{BallsFaced} + \beta_2 \cdot \text{Position} + \beta_3 \cdot \text{Experience} + \varepsilon_{it}$$

α_i is the batter's unobserved individual effect — his innate talent/temperament. Estimating it correctly is the whole challenge.

THE PANEL AT A GLANCE

410

players (n)

19

seasons (T)

1,676

player-seasons

Unbal.

entry / exit

Unbalanced: players debut & retire, so selection/attrition must be kept in mind.

Takeaway Panel structure lets us hold each batter's fixed quality constant and ask what really moves performance.

Free, public, ball-by-ball — fully reproducible

Nothing proprietary and nothing from Kaggle: every variable is engineered from raw deliveries.

① Match data — Cricsheet

Every IPL delivery (2008–2026) as CSV. We compute runs, balls, batting position, opposition and venue ourselves — so the entire pipeline can be rerun from scratch.

② Age — exact-ID join (no scraping)

Cricsheet has no birth dates, so we link each player: Cricsheet register → ESPNcricinfo ID → Wikidata date of birth. This filled 98% of players automatically.

The raw material

295,557

deliveries

1,243

matches

19

seasons

98%

ages auto-filled

Takeaway *A transparent, free dataset — a reviewer can reproduce every number from the public source files.*

From 296k deliveries to one player-season panel

Each delivery is aggregated to a single row per batter per season, adding role and match-context measures.

Variable	What it captures (engineered from ball-by-ball)
runs · balls · strike_rate	season output: runs, legal balls faced, and 100-runs/balls
position	mean batting order (where in the line-up he bats)
experience	cumulative seasons played (career stage)
opp_econ	avg economy of bowling sides faced — quality of opposition
venue_rpo	avg run-rate of grounds batted at — how batting-friendly
pp_share · death_share	share of balls in powerplay / death overs — his role
lag_balls · lag_position · lag_sr	previous qualifying season — predetermined, used to fight reverse causality
age · age ²	season-year – birth-year, and its square (for the aging curve)

Top block = the variables in our core model. Lower block = controls (context, role) and predetermined lags that make the comparison fair.

Takeaway Context controls let us compare like with like; lags pre-empt the 'good players get more balls' problem.

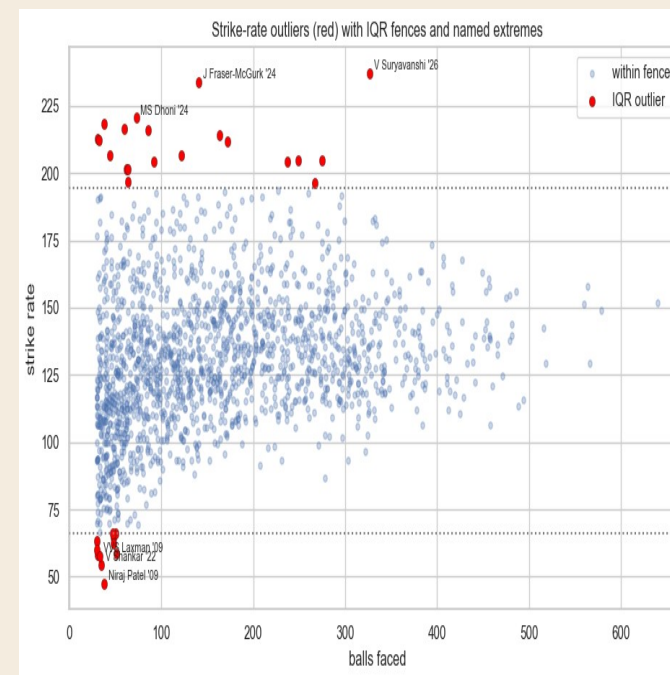
Missing data, the exposure floor & outliers

Before modelling we clean sensibly: handle gaps, set a minimum exposure, and look at the extremes.

Missing data is by design — a debut season has no 'last season', and 2% of ages are unmatched — so we drop incomplete rows per model.

The exposure floor: strike rate is wild off a handful of balls, so we require ≥ 30 balls per season. Remaining extremes are real cricket, so we keep them.

Robustness to this 30-ball choice is shown later (slide 13).

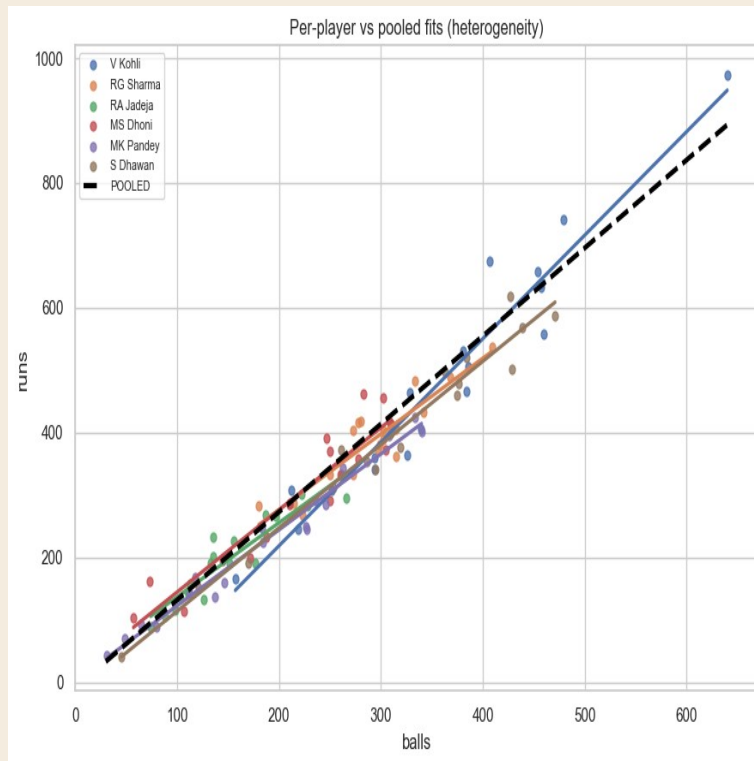


Each dot = a player-season; red = statistical (IQR) outliers. The top strike rates are death-over finishers (Dhoni, 95% death balls) and explosive openers — signal, not error.

Takeaway *Cleaning removes noise, not information: the 'outliers' are exactly the players the model later explains.*

Do batters differ enough to need a panel?

If batters were alike, one pooled line would fit everyone. They are not — so we need individual effects.



Each colour is one batter's own balls→runs line; the black dashed 'pooled' line fits none of them well.

Between- vs within-batter variation

How much of the spread is across different players vs. within one player over time?

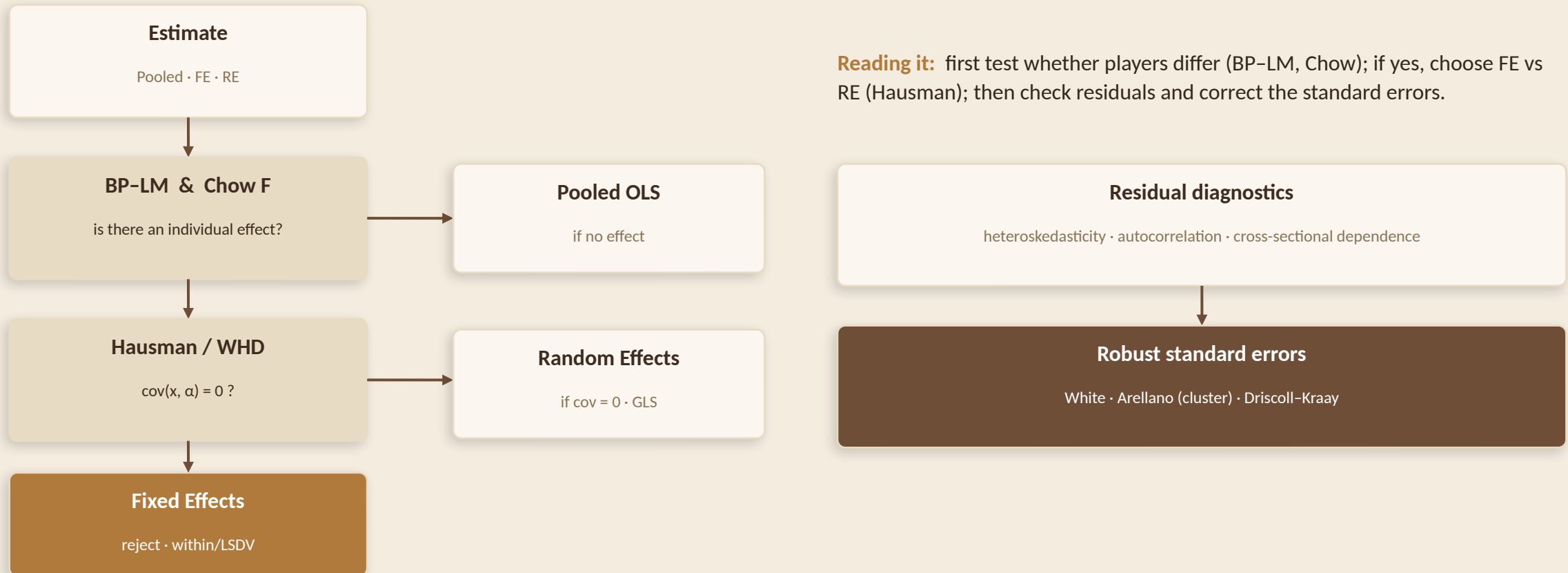
	overall	between	within
runs	162	112	110
strike rate	26	24	19
position	2.1	2.2	0.8
experience	3.8	2.2	2.8

Big 'between' values = a strong α_i . Pooling forces one slope on everyone → heterogeneity bias. FE / RE fix this.

Takeaway Clear player-to-player differences justify Fixed/Random Effects over a single pooled regression.

How we choose the model — the decision tree

A standard panel decision tree picks the estimator; we run every test it calls for.



Reading it: first test whether players differ (BP-LM, Chow); if yes, choose FE vs RE (Hausman); then check residuals and correct the standard errors.

Takeaway Our data: reject pooling (BP-LM & Chow) → reject RE (Hausman) → Fixed Effects → clustered SEs.

The obvious model, and why it misleads

We start by explaining season runs, climbing the Pooled → FE → RE ladder and testing which fits.

coef[sig]	Pooled	FE	RE
balls	1.42***	1.44***	1.43***
position	1.23*	1.96	1.01*
experience	0.79**	1.71***	1.13***

All three agree balls drives runs; they disagree on experience — an early sign pooling mishandles player differences. (***) $p < .01$ (**) $p < .05$ (*) $p < .10$, cluster-robust SE.)

Which model? The tests decide

- Chow $F = 2.34$, $p < 1e-15$ → reject pooling
- Breusch-Pagan LM = 372.8, $p < 1e-80$ → an individual effect exists
- Hausman $\chi^2 = 26.18$, $p < 0.0001$ → use Fixed Effects

The red flag

0.93

within- R^2 of the FE runs model — suspiciously high.

Runs $\approx$ Balls $\times$ StrikeRate $\div 100$. **Regressing runs on balls is almost an accounting identity — we're explaining arithmetic, not skill.**

Also diagnosed: reverse causality (better players get more balls) and selection. → Stage 2.

Takeaway The tests clearly pick Fixed Effects — but the model itself is mechanical, so we must reframe the question.

Model quality (strike rate), not quantity

Four changes turn the mechanical model into one that explains skill and match context.

Dependent var → strike rate

removes the balls ↔ runs identity

Lag balls & position

predetermined → breaks reverse causality

Add opposition • venue • over-phase

controls for match context

Add season fixed effects

absorbs league-wide scoring trends

Full strike-rate panel — coefficient [significance]

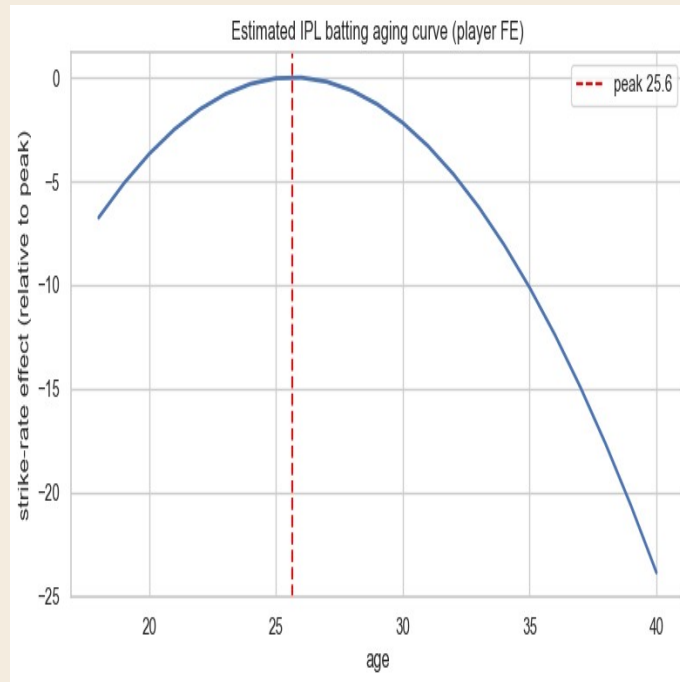
variable	Pooled OLS	FE (preferred)
death_share	27.8***	42.9***
venue_rpo	8.5***	16.0***
opp_econ	9.7***	12.6**
lag_position	-4.3***	-1.7**
lag_balls	0.00	-0.02**
pp_share	-11.9**	-6.0
experience	-0.4*	-1.3

Death-overs & venue are large and significant in both columns; pooling over/understates the rest by ignoring who is batting. (Full Pooled/FE/RE + SEs in the notebook.)

within-R² 0.93 → 0.17: the mechanical fit is gone — we now explain skill & context, not arithmetic. Hausman $\chi^2=59.0$ → Fixed Effects again.

When do batters peak — and a caveat

Adding Age and Age² lets the effect of age bend, revealing a peak — but with an identification limit.



Estimated effect of age on strike rate (player fixed effects). The curve rises, peaks, then falls.

age +5.94 (p=0.002)
 age² -0.116 (p<0.001)
 → inverted-U, peak ≈ 25.6 years

The age–period–cohort trap

Age, experience and the calendar all move +1 each season, so they cannot be told apart at once.

Adding season effects (Spec C) FAILS — age is perfectly absorbed.

We identify the curve in Spec A and report the failure honestly; the peak (curvature) is the robust part.

Takeaway Strike rate peaks in the mid-20s; we're transparent that age can't be fully separated from era & experience.

Does good form carry over to next season?

Adding last season's strike rate tests persistence; a lagged outcome biases FE, so we use Arellano–Bond GMM.

The Nickell-bias bracket — three estimates of persistence (ρ on lagged strike rate)

−0.08

ρ (Fixed Effects) · biased DOWN

+0.16

ρ (Arellano–Bond) · consistent

+0.21

ρ (Pooled OLS) · biased UP

FE under-states and OLS over-states persistence; Arellano–Bond GMM sits between them — exactly as theory predicts.

Is the GMM valid?

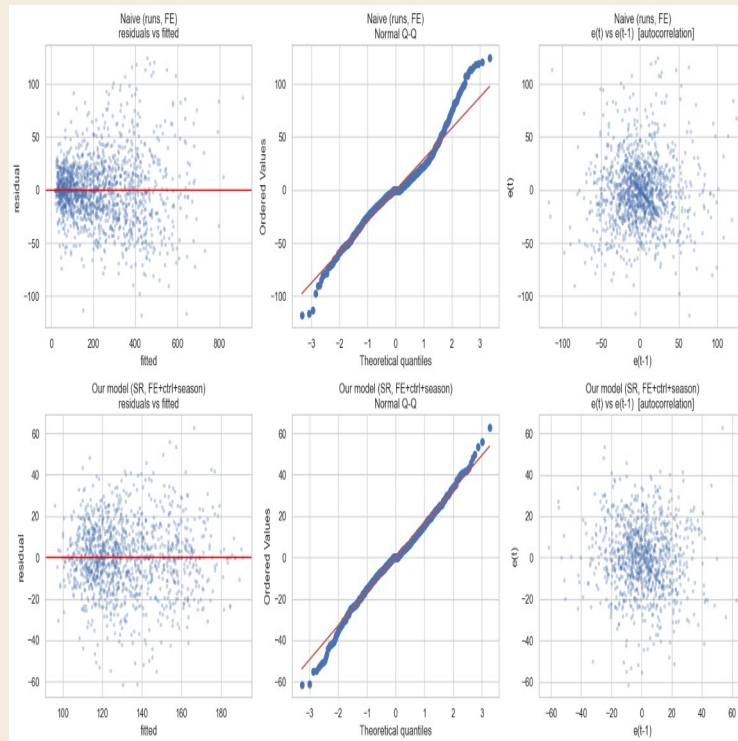
- Hansen J: $\chi^2=0.29$, $p=0.87$ → instruments valid ✓
- AR(1): $p<0.001$ ✓ AR(2): $p=0.011$ (borderline)
- $\rho(AB)=0.16$, $p=0.026$ — only small persistence

What it means

Persistence is weak — a batter's strike rate barely carries over once bias is removed; form is largely re-earned each season. **But death-overs & venue stay strong in every estimator.**

Are the residuals well-behaved?

We interrogate the residuals of the naive vs final model — heteroskedasticity, normality, autocorrelation.



Top row = naive model, bottom = final. Tighter, more centred clouds and a straighter Q-Q line mean the final model's residuals are far better behaved.

Test statistics (lower = healthier)

test	naive	ours
Breusch-Pagan	221	56
Jarque-Bera	397	41
AR(1) ρ	0.04	-0.07

Stats fall 4–10× from naive to final — much healthier, though still mildly heteroskedastic. That is exactly why we report cluster-robust SEs throughout; the coefficient estimates do not change.

Takeaway Diagnostics confirm the final model is sound; remaining imperfections are absorbed by robust standard errors.

Cross-sectional dependence · SEs · thresholds

Three final checks: do players co-move, do robust SEs change anything, and does the ball-floor matter?

1 · Cross-sectional dependence (Pesaran CD)

- no season FE: CD = -0.30, p = 0.77
 - with season FE: CD = -1.09, p = 0.28
- players don't co-move; clustered SEs suffice.

2 · Driscoll–Kraay SEs (belt-and-braces)

coef	SE clustered	SE D-K
death_share	7.8	3.5
venue_rpo	3.9	3.6

Significance is unchanged under either standard-error choice.

3 · Robustness to the exposure floor (MIN_BALLS)

Re-estimating the model with the floor set anywhere from 10 to 100 balls: **the death-over (+28 to +43) and venue (+11 to +16) effects stay positive and significant** — only statistical power dips at very high cut-offs. The 30-ball floor sits in the stable mid-range, so no conclusion depends on it.

WHAT WE FOUND

What drives the strike rate

Effects on strike rate from the preferred fixed-effects model (all highly significant):

+43

death-over share

batting at the death — the biggest lever

+16

venue run-rate

high-scoring grounds lift scoring

+13

opposition economy

weaker bowling → faster scoring

Aging: strike rate peaks at ≈ 26 , then gently declines

Persistence: weak — last season's form barely predicts this season's

Why fixed effects: Chow + BP-LM reject pooling; Hausman rejects RE

Headline: output is shaped by role & match context, not by stable individual 'form'

Honest limits, and where this goes next

Good empirical work names its weaknesses — here is what we solved, what we acknowledge, and what's next.

Caveats — solved or acknowledged

- Mechanical identity → solved (use strike rate)
- Time-invariant talent → solved (fixed effects)
- Lagged-outcome bias → solved (Arellano-Bond)
- Reverse causality → mitigated (lagged regressors)
- Role/phase mildly endogenous → acknowledged
- Survivorship / selection → acknowledged
- Age-period-cohort → identified under a restriction

Further research

- System GMM (Blundell-Bond); tune lags for AR(2)
- Heckman selection model for survivorship
- Pay-performance: auction prices with IV (Session 3)
- Richer context: bowler quality, match situation
- Quantile panel: anchors vs aggressors
- Other outcomes: boundary %, dismissal rate

Thank you

Questions & discussion

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